

Actividad Video #1

$$a) \begin{cases} x_1' = -x_1 \\ x_2' = 3x_2 \end{cases}, \quad \begin{matrix} x_1(0) = 1 \\ x_2(0) = -2 \end{matrix}$$

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix}$$

$$\lambda = -1$$

$$\lambda = 3$$

$$\begin{aligned} x_1(t) &= c_1 e^{-t} \\ x_2(t) &= c_2 e^{3t} \end{aligned}$$

$$\Phi(t) = \begin{bmatrix} e^{-t} & 0 \\ 0 & e^{3t} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

Eliminar Parametro t

$$\Phi(t) = \begin{bmatrix} c_1 e^{-t} \\ c_2 e^{3t} \end{bmatrix}; t \in \mathbb{R}$$

$$x_1 = c_1 e^{-t}$$

$$x_2 = c_2 e^{3t}$$

$$x_2 = c_2 (e^t)^3$$

$$x_2 = c_2 \left(\frac{c_1}{x_1} \right)^3$$

$$x_1 = \frac{c_1}{e^t}$$

$$e^t = \frac{c_1}{x_1}$$

$$x_2 = c_2 \cdot \frac{c_1^3}{x_1^3} = \frac{C}{x_1^3}$$

$$x_1^3 \cdot x_2 = C \quad (C \in \mathbb{R})$$

$$x_2 = \frac{C}{x_1^3} \quad (x_1 \neq 0)$$

Reemplazando V. Iniciales

$$1 = c_1$$

$$-2 = c_2$$

$$\phi(t) = \begin{bmatrix} 1e^{-t} \\ -2e^{3t} \end{bmatrix}$$

$$x_1 = 1e^{-(0)} = 1$$

$$x_2 = -2e^{3(0)} = -2$$

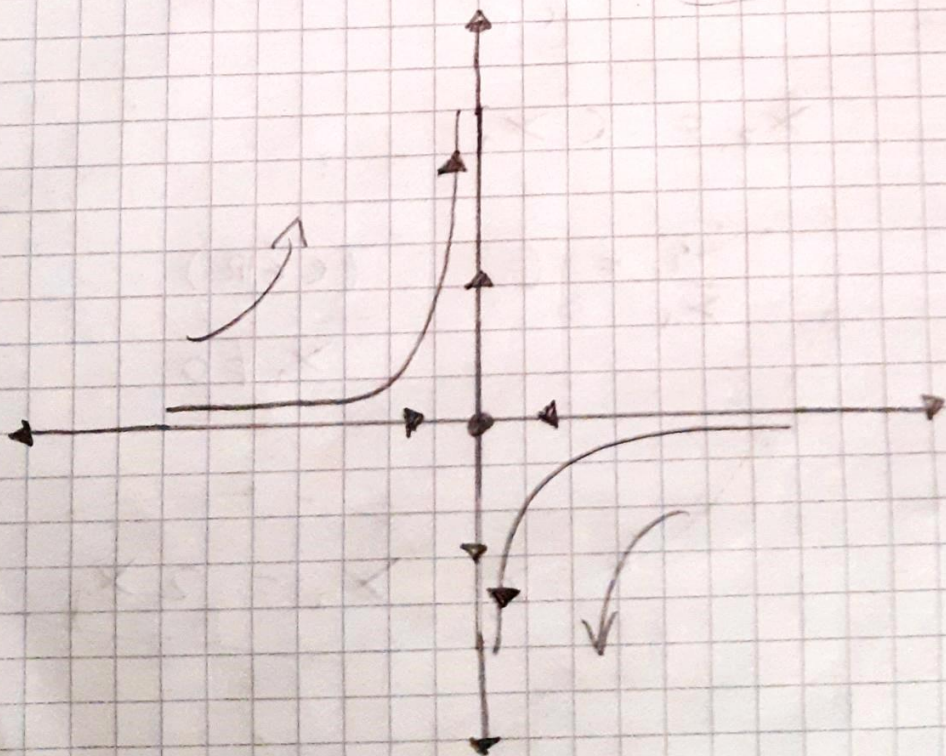
$$x_1^3 \cdot x_2 = c_1 \cdot c_2$$

$$(1)^3(-2) = (1)(-2)$$

$$-2 = -2$$

Tipo Silla

$$x_2 = \frac{-2}{x_1^3}$$



$$x_1 = c_1 e^{-t} \rightarrow 0$$

$$x_2 = c_2 e^{3t} \rightarrow \infty$$

b) $\begin{cases} x_1' = 2x_1 \\ x_2' = 2x_2 \end{cases} \quad \begin{matrix} x_1(0) = 1 \\ x_2(0) = 2 \end{matrix}$

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \rightarrow$$

$$\lambda_1 = 2$$

$$\lambda_2 = 2$$

$$\Phi(t) = \begin{pmatrix} e^{2t} & 0 \\ 0 & e^{2t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

$$= \begin{pmatrix} c_1 e^{2t} \\ c_2 e^{2t} \end{pmatrix}$$

$$x_1(t) = c_1 e^{2t}$$

$$x_2(t) = c_2 e^{2t}$$

$$x_1 = c_1 e^{2t}$$

$$x_2 = c_2 e^{2t}$$

$$e^{2t} = \frac{x_1}{c_1}$$

$$x_2 = c_2 \frac{x_1}{c_1}$$

$$x_2 = c x_1$$

$$\frac{x_2}{x_1} = c \quad (c \in \mathbb{R})$$

$$x_1 \neq 0$$

PVZ

$$c_1 = 1$$

$$c_2 = 2$$

$$x_1 = 1$$

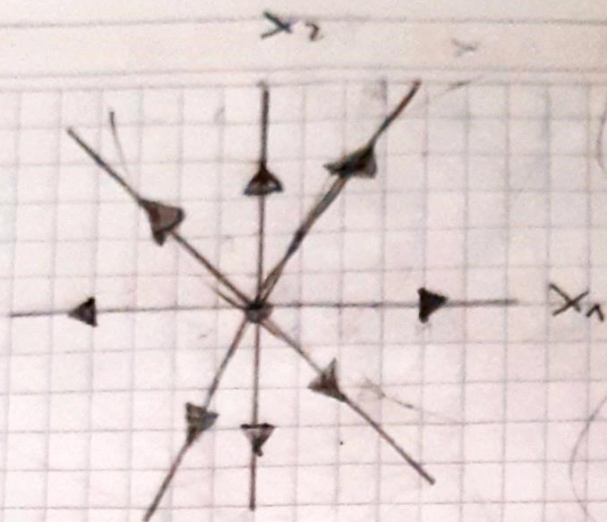
$$x_2 = 2$$

$$\frac{2}{1} = c$$

$$2 = c$$

$$x_2 = 2x_1$$

Nodo
Estrella



• Órbita sobre x_2

$$c_1 = 0$$

$$\begin{pmatrix} 0 \\ c_2 e^{2t} \end{pmatrix}$$

• Órbita sobre x_1

$$c_2 = 0$$

$$\begin{pmatrix} c_1 e^{2t} \\ 0 \end{pmatrix}$$

$$x_1' = -3x_1 - x_2$$

$$x_2' = x_1 - 3x_2$$

$$x_1(0) = 1$$

$$x_2(0) = 1/2$$

$$A = \begin{pmatrix} -3 & -1 \\ 1 & -3 \end{pmatrix}$$

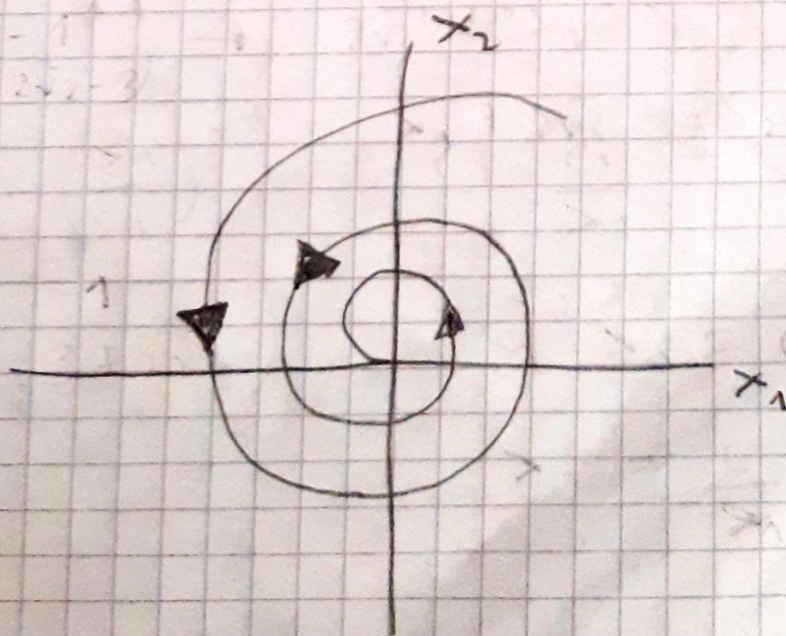
$$\det = a - (m)$$

$$a + 1 = 10$$

$$(a) (b)$$

$b > 0$ sentido Antihorario

$a = -3$, las soluciones tenderán a Cero



Video #2

$$\begin{aligned} a) \quad Y_1' &= \frac{1}{4} Y_1 + Y_2 \\ Y_2' &= Y_1 + \frac{1}{4} Y_2 \end{aligned}$$

$$\tau = 1/2 > 0$$

$$\begin{pmatrix} 1/4 & 1 \\ 1 & 1/4 \end{pmatrix}$$

$$\begin{aligned} d &= 1/16 - 1 \\ &= -\frac{15}{16} < 0 \end{aligned}$$

$$\lambda_{1,2} = \frac{\tau \pm \sqrt{\tau^2 - 4d}}{2}$$

$$= \frac{1/2 \pm \sqrt{1/4 - 4(-15/16)}}{2}$$

$$\lambda_1 = \frac{1/2 + 2}{2} = 5/4$$

$$= \frac{1/2 \pm \sqrt{4}}{2} = \frac{1/2 \pm 2}{2}$$

$$\lambda_2 = \frac{1/2 - 2}{2} = -3/4$$

$$(\lambda - 3/4)$$

$$\begin{pmatrix} -1 & 4 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$V_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$-u - v = 0 \quad \Rightarrow \quad v = -u$$

$$u = -v$$

$$\lambda = 5/4$$

$$\begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$P^{-1}AP = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1/4 & 1 \\ 1 & 1/4 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

$$J = \begin{pmatrix} 5/4 & 0 \\ 0 & -3/4 \end{pmatrix}$$

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = J \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Sistema
Canónico

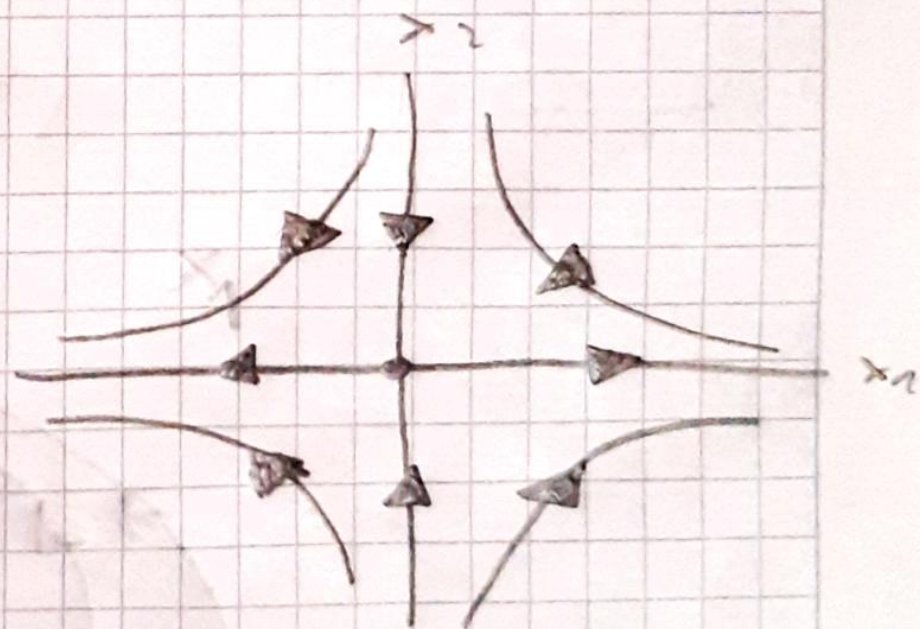
$$u - v = 0$$

$$u = v$$

$$v = v$$

$$V_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$X(t) = \begin{pmatrix} c_1 e^{5/4 t} \\ c_2 e^{-3/4 t} \end{pmatrix}$$



$$b) \quad x' = -3x$$

$$y' = -\frac{1}{10}y$$

$$\begin{pmatrix} -\lambda - 3 & 0 \\ 0 & -\lambda - \frac{1}{10} \end{pmatrix}$$

$$\begin{pmatrix} -3 & 0 \\ 0 & -1/10 \end{pmatrix}$$

det

$$= \frac{3}{10} > 0$$

$$\lambda_1 = -3$$

$$\lambda_2 = -1/10$$

$$\tau = -3 - 1/10$$

$$= -\frac{31}{10}$$

$$\Delta = \tau^2 - 4d$$

$$= \left(-\frac{31}{10}\right)^2 - 4\left(\frac{3}{10}\right) = 8,41$$

$$\lambda = -3 \quad \begin{pmatrix} 0 & 0 \\ 0 & \frac{29}{10} \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 0 \\ 0 & 29 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$v = 0$$

$$u = u$$

$$V_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\lambda = -1/10$$

$$\begin{pmatrix} -\frac{29}{10} & 0 \\ 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} -29 & 0 \\ 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$u = 0$$

$$v = v$$

$$V_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$Y_1 = C_1 e^{-3t}$$

$$Y_2 = C_2 e^{-1/10 t}$$

$$-3 < \lambda_1 < \lambda_2 < 0$$

como ambos son negativos,

la grafica sera tipo

Nodo "Sumidero"

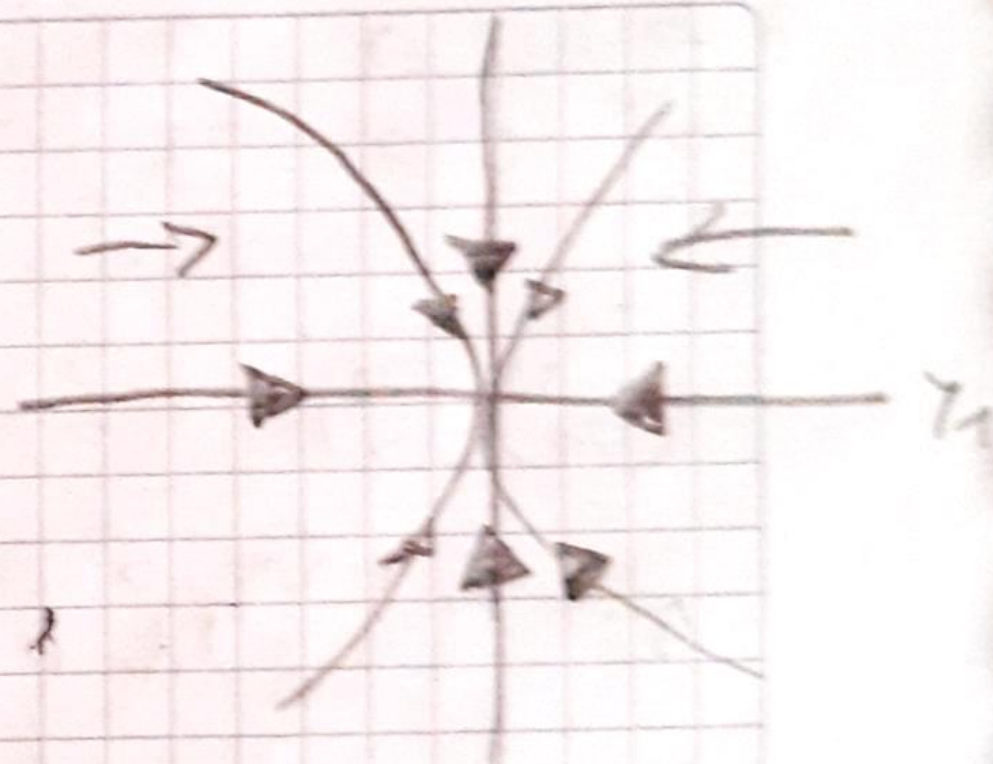
$$Y' = \begin{pmatrix} -2.5 & 0 \\ 0 & -2 \end{pmatrix} Y$$

$$\Delta = 5 > 0$$

$$\tau = -4.5$$

$$\lambda = -2.5$$

$$\lambda = -2$$



$$\Delta = (-4.5)^2 - 4(5)$$

$$= 20.25 - 20 = 0.25$$

$$\lambda = -40, 25$$

$$\begin{aligned} c) \quad x' &= 2x + y \\ y' &= 2y \end{aligned}$$

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} \quad \det = 4 \neq 0$$

$$\lambda_1 = 2$$

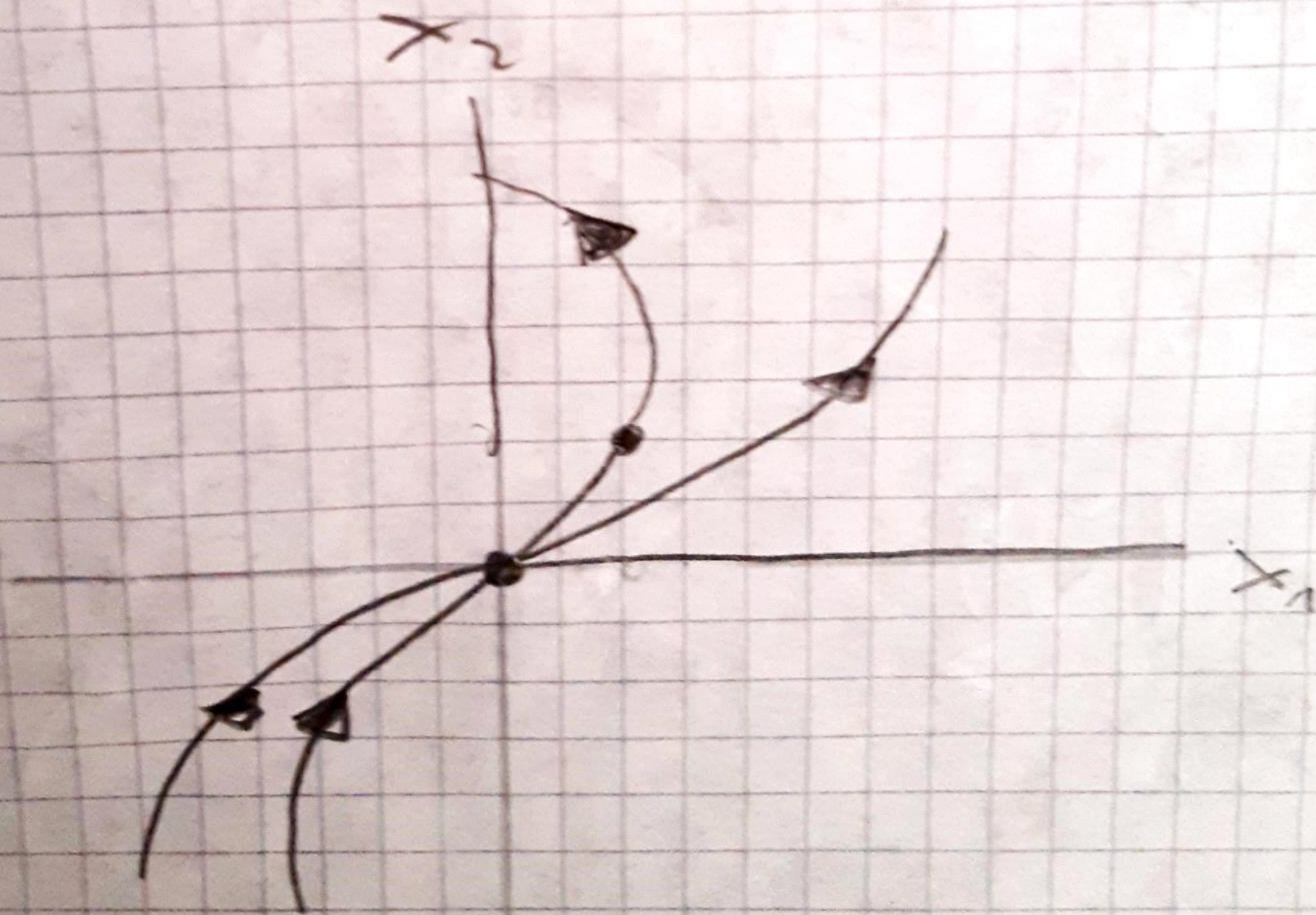
$$\lambda_2 = 2$$

$$\lambda_1 = \lambda_2 > 0$$

forma canônica de

Nodo Impróprio

$$\begin{cases} x_1 = c_1 e^{2t} + c_2 t e^{2t} \\ x_2 = c_2 e^{2t} \end{cases}$$



Video #3

$$I) \quad \dot{x} = \begin{pmatrix} 2 & -3 \\ 4 & 1 \end{pmatrix} x$$

$$\tau = 3$$

$$\det = 2 - (-12) = 14$$

$$\Delta = 9 - 4(14)$$

$$= 9 - 56$$

$$= -47 \text{ for } \tau \text{ unstable}$$

$$\lambda_{1,2} = \frac{\tau \pm \sqrt{\tau^2 - 4d}}{2}$$

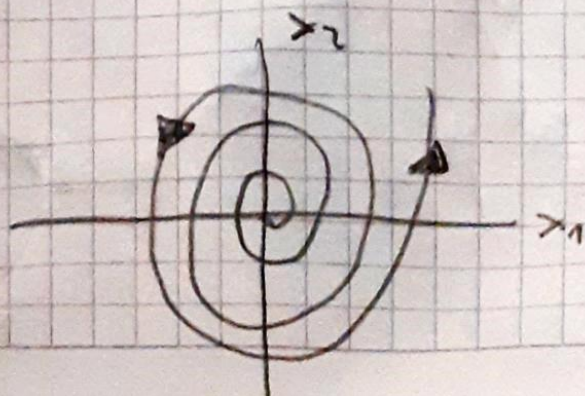
$$= \frac{3 \pm \sqrt{-47}}{2}$$

$$= \frac{3 \pm \sqrt{47}i}{2}$$

$$V_1 = \begin{pmatrix} \frac{\sqrt{47}i}{8} + 1 \\ 8 \\ 1 \end{pmatrix} = \begin{pmatrix} 1/8 \\ 1 \end{pmatrix} + \begin{pmatrix} \sqrt{47}i/8 \\ 0 \end{pmatrix}$$

IR Imaginario

$$P = \begin{pmatrix} \sqrt{47}/8 & 1/8 \\ 0 & 1 \end{pmatrix} \quad \det P > 0$$



$$\text{III)} \quad x' = \begin{pmatrix} 2 & 2 \\ -6 & -4 \end{pmatrix} x$$

$$\Delta = 4 - 4(4)$$

$$\tau = -2$$

Estable

$$\det = -8 - (-12)$$

$$= 4$$

$$= -12$$

foco

$$\lambda_{1,2} = \frac{\tau \pm \sqrt{\tau^2 - 4\det}}{2} = \frac{-2 \pm \sqrt{12}}{2}$$

$$= -1 \pm \sqrt{3}i$$



$$\text{II)} \quad x' = \begin{pmatrix} 0 & -3 \\ 2 & 2 \end{pmatrix} x$$

$$\tau = 2$$

$$\det = 6$$

$$\Delta = -20$$

foco

inestable

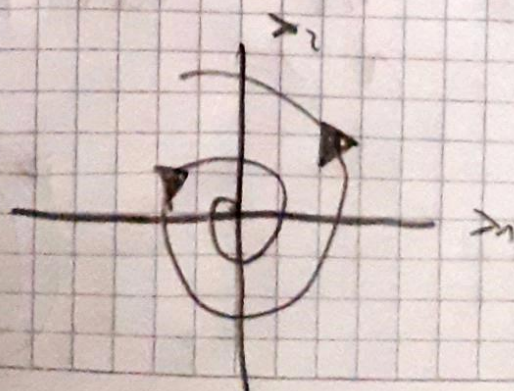
$$\lambda_{1,2} = \frac{\tau \pm \sqrt{\tau^2 - 4\det}}{2}$$

$$= 1 \pm \sqrt{5}i$$

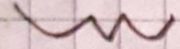
$$P = \begin{pmatrix} \sqrt{5}/2 & -1/2 \\ 0 & 1 \end{pmatrix}$$

$$\det P > 0$$

$$w = \begin{pmatrix} \frac{\sqrt{5}i - 1}{2} \\ 1 \end{pmatrix}$$



$$\text{IV} \quad X' = \begin{pmatrix} 2 & -3 \\ 3 & -2 \end{pmatrix} X$$

$\zeta = 0$

 Tied center

$$\det = -4 - (-9) = 5$$

$$\Delta = -4(5)$$

$$= \begin{pmatrix} -20 & 10 \\ 10 & -20 \end{pmatrix}$$

$$\lambda_{1,2} = \frac{\pm \sqrt{-20}}{2} = \pm \sqrt{5} i$$

$$V_1 = \begin{pmatrix} \frac{\sqrt{5}i + 2}{3} \\ 1 \end{pmatrix}, \quad P = \begin{pmatrix} \sqrt{5}/3 & 2/3 \\ 0 & 1 \end{pmatrix}$$

$$\det P \neq 0$$

$$v) \quad X' = \begin{pmatrix} -1 & 1 \\ -3 & 1 \end{pmatrix} X$$

$$\lambda = 0$$

$$\det = -1 - (-3)$$

$$= 2$$

$$\Delta = -4(2)$$

$$= -8$$

$$\lambda_{1,2} = \frac{\lambda \pm \sqrt{\lambda^2 - 4d}}{2} = \frac{\pm \sqrt{-8}}{2}$$

$$= \frac{\pm \sqrt{8}i}{2}$$

$$\lambda_1 = \frac{\sqrt{8}i}{2} \quad \lambda_2 = -\frac{\sqrt{8}i}{2}$$

$$= \frac{2\sqrt{2}i}{2} = -\frac{2\sqrt{2}i}{2}$$

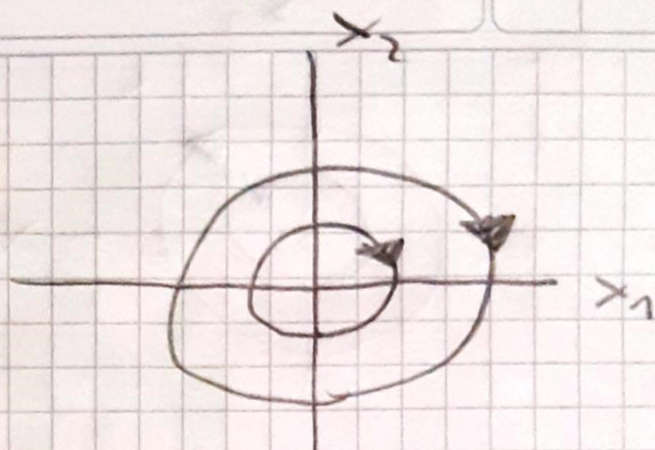
$$= \sqrt{2}i = -\sqrt{2}i$$

$$V_1 = \begin{pmatrix} \frac{-\sqrt{2}i + 1}{3} \\ 1 \end{pmatrix} = \begin{pmatrix} 1/3 \\ 1 \end{pmatrix} + \begin{pmatrix} -2\sqrt{3} \\ 0 \end{pmatrix}$$

Real Imaginary

$$P = \begin{pmatrix} -\sqrt{2}/3 & 1/3 \\ 0 & 1 \end{pmatrix}$$

$$\det P < 0$$



VII $x' = \begin{pmatrix} 1 & 2 \\ -5 & -3 \end{pmatrix} x$

$$\lambda = -2$$

Stable

$$\det = -3 - (-10) = 7 > 0$$

$$\Delta = 4 - 4(7)$$

$$= -24 < 0$$

$$F = 0$$

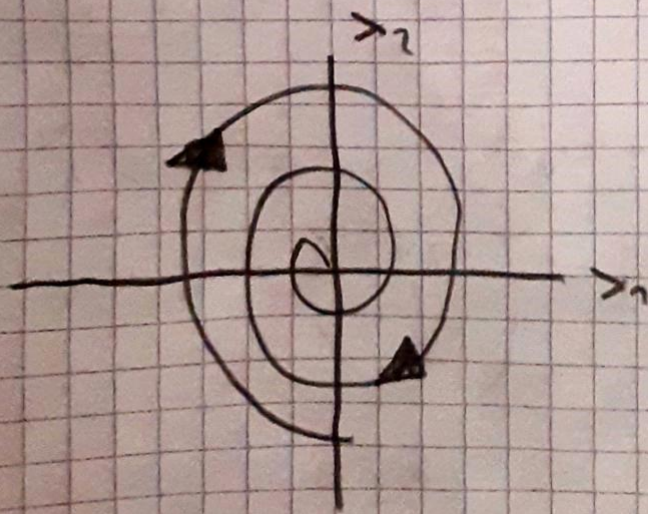
$$\lambda_{1,2} = \frac{-2 \pm \sqrt{-24}}{2}$$

$$= -1 \pm \sqrt{6}i$$

$$V_1 = \begin{pmatrix} \frac{-\sqrt{6}i - 1}{5} \\ 1 \end{pmatrix}$$

$$P = \begin{pmatrix} -\sqrt{6}/5 & -2/5 \\ 0 & 1 \end{pmatrix}$$

$$\det P < 0$$



$$\bullet \quad \frac{dx}{dt} = \begin{pmatrix} a & b \\ b & a \end{pmatrix} x$$

$$\lambda = 2a \quad \det = a^2 - b^2$$

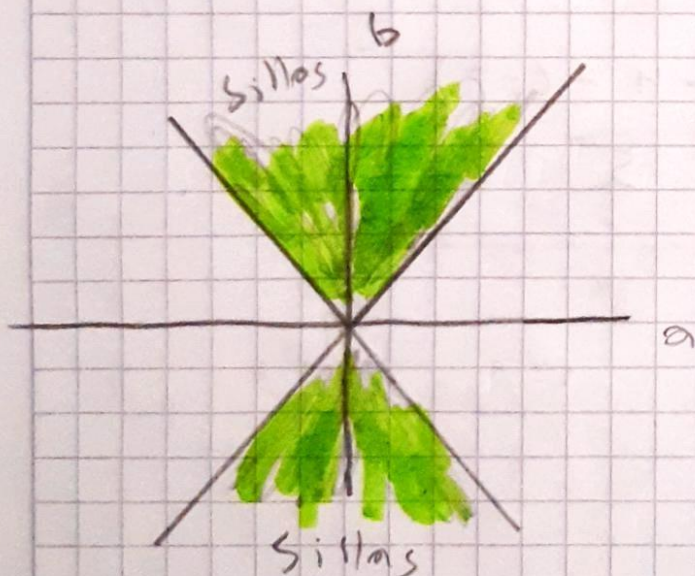
$$\begin{aligned} \Delta &= 4a^2 - 4(a^2 - b^2) \\ &= \cancel{4a^2} - \cancel{4a^2} + 4b^2 \\ &= 4b^2 > 0 \end{aligned}$$

$$\sqrt{a^2} = \sqrt{b^2}$$

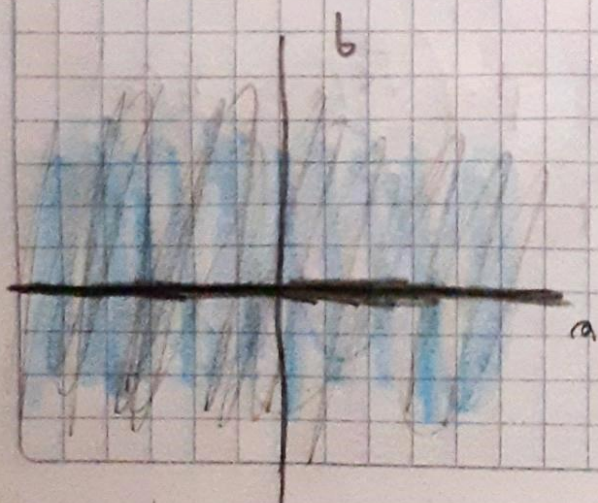
$$a = \pm b$$

$$a^2 \neq (3)^2 < 0$$

$$a^2 = (-3)^2 < 0$$

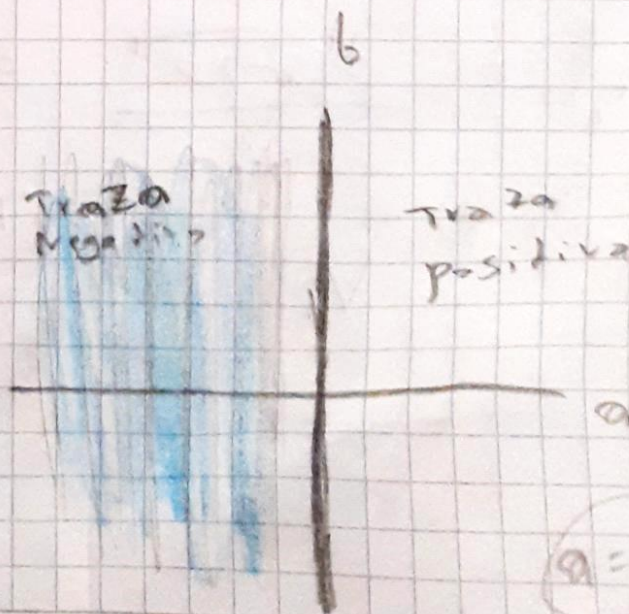


Det. Positivo



Trazo
Negativo

Trazo
positiva



$$b=0$$

$$a=0$$

Nodo
Estable

Sillas

Sillas

Nodo
Inestable